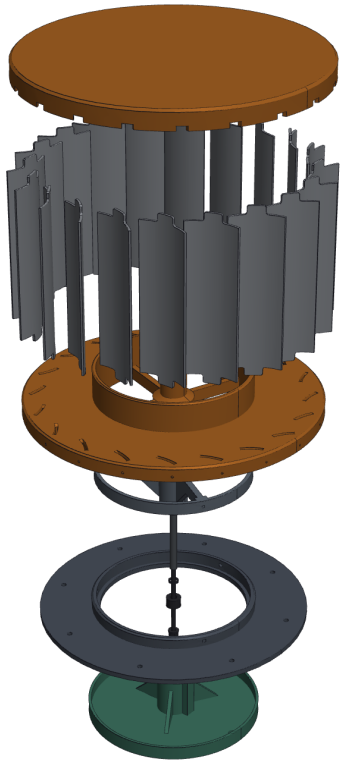
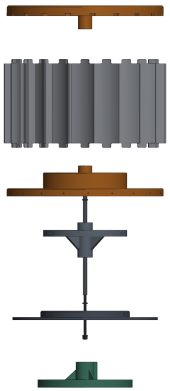
SHEET 4 - EXPLODED ASSEMBLY (true CAD) AND ASSEMBLY ORDER



ASSEMBLY ORDER (proved collision-free in CAD, gate 10):

1 Press both R4-2RS bearings into the support tower (top seat from above, bottom seat from below) - fit chosen from the fit coupon.
2 Bottom washer + 1/4 in shaft collar loosely on the shaft; slide shaft UP through both bearings from below.
3 Top thrust washer onto the shaft, on the upper inner race.
4 Lower hub down the shaft until its cup ceiling sits on the washer. Mark through the set-screw hole; spot-drill dimple (guide); M4 set screw + nut, threadlocker. Set collar for 0.2 mm endplay.
5 Drop each blade's lower tab straight DOWN into a lower-hub pocket (all 20 the same way: concave side as in the plan view).
6 Captive M4 nut into the upper hub side slot. Lower the hub straight DOWN over all upper tabs (lead-ins align them).
7 M4 rod up through the tube into the nut; washer + nyloc from below; snug until every shoulder seats. Blades trapped, no blade hardware.
8 Spin by hand: no rub, no rattle. Static-balance (tape on hub rim).
9 Fasten roof adapter (sealant + 8 screws); drop the rotor cartridge into the collar; 3x M3 radial screws. Optional tray underneath.

Exploded offsets are along +Z only, matching the real assembly direction: blades enter the lower hub from ABOVE; the upper hub comes DOWN over the upper tabs. No part has to be threaded over an already-bent blade.



DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Exploded assembly

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

SHEET 1/1 Source: build123d parametric model + MASTER_PARAMETERS.json